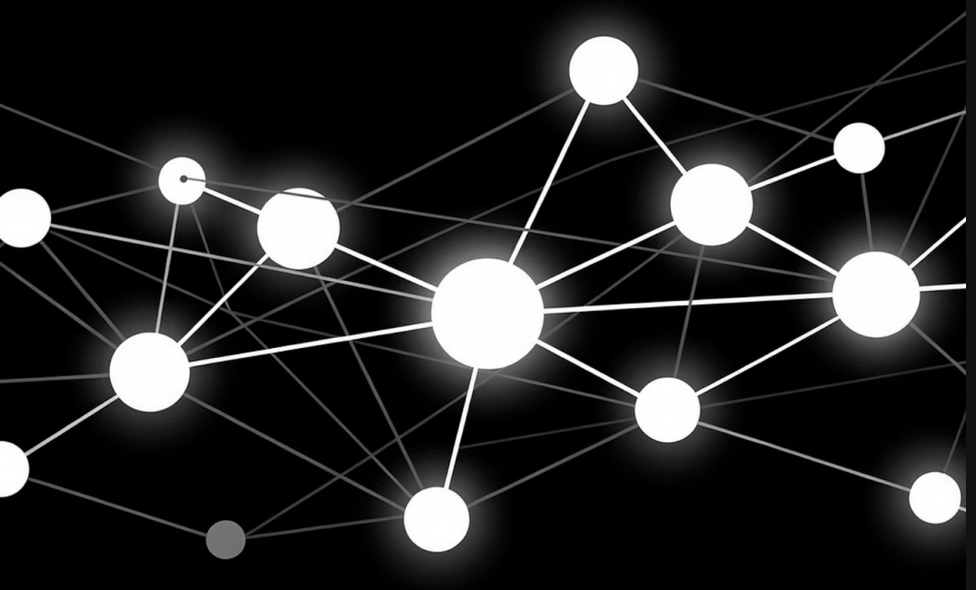

Graph-Constrained Vector Retrieval

for Mitigating Structural Failure Modes in Naive RAG

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QASPER DATASET | LLAMA 3.2 3B | NETWORKX GRAPH | RAGAS EVALUATION

github.com/bharathparimanan/rag-structural-eval

Presentation - March 2026

Introduction - Motivation and Problem Statement

RAG has become the dominant architecture for grounding LLM responses in external knowledge. Naive RAG is the most widely deployed pattern - simple, fast, requiring no retraining. But that simplicity comes at a structural cost: two aggressive simplifications discard information available in the source documents.

Context Loss

Fixed-window chunking is topologically blind - it severs paragraph boundaries and narrative continuity. When evidence splits across two fragments, neither alone is sufficient. The evidence exists in the corpus; it simply cannot be retrieved as a coherent unit.

MEASURED BY RAGAS CONTEXT RECALL

Retrieval Noise

Pure cosine similarity measures vector proximity, not evidential relevance. When topically similar but evidently irrelevant chunks are surfaced, the LLM generates an answer that *appears* grounded but is built on the wrong evidence - a distracting false positive.

MEASURED BY RAGAS CONTEXT PRECISION

Threshold: Both failure modes are operationalised with a hard threshold - below 0.70 = failure. By fixing both, this project achieves structurally coherent, traceable, evidence-grounded RAG without full knowledge graph overhead.

Objectives and Research Question

> To what extent do graph-based structural constraints on vector retrieval mitigate Context Loss and Retrieval Noise in Naive RAG pipelines, as measured by RAGAS Context Recall, Context Precision, Faithfulness, and Provenance Coverage on QASPER?

SQ1 - Context Loss

Does paragraph-boundary chunking alone (V1) measurably reduce Context Loss versus fixed-window chunking (V0)?

SQ2 - Retrieval Noise

Does the graph proximity filter (V2) further reduce Retrieval Noise beyond paragraph chunking alone (V1)?

SQ3 - Provenance

Does graph-constrained retrieval enable traceable answer generation, measured by Provenance Coverage?

Contribution framing: This is not a system-building project - the contribution is controlled experimental evidence. Graph RAG (Edge et al. 2024) requires knowledge extraction, entity resolution, and LLM calls at indexing time. This project tests a lighter hypothesis: structural constraints at chunking and retrieval stages alone are sufficient to mitigate both failure modes. Provenance Coverage operationalises explainability as a computable metric.

Related Work

Gao et al. (2023) - RAG Survey

Defines Naive, Advanced, and Modular RAG paradigms; establishes the seven failure mode taxonomy including Context Loss and Retrieval Noise. *arXiv:2312.10997*

Differs: This project targets exactly two failure modes and provides failure-mode-disaggregated evaluation - not aggregate accuracy.

Wang et al. (2023) - FILCO

Trains a context filtering model to remove irrelevant passages before generation using lexical and information-theoretic scoring. *arXiv:2311.08377*

Differs: FILCO requires a separate filter model. This project applies structural graph constraints - no training, model-agnostic, lighter intervention.

Jimeno-Yepes et al. (2024) - Chunking

Proposes structure-aware chunking by document element type, showing paragraph-level chunking ignores structural information and reduces retrieval quality. *arXiv:2402.05131*

Differs: Directly motivates V1. This project extends to graph-constrained retrieval (V2) and evaluates on a scientific QA benchmark.

Peng et al. (2024) - GraphRAG Survey

Comprehensive survey of GraphRAG systems showing graph-based retrieval captures relational knowledge unattainable with flat dense embeddings. *arXiv:2408.08921*

Differs: GraphRAG is the heavyweight solution. This project tests whether lighter structural constraints are sufficient - without entity extraction overhead.

Dataset - QASPER and Ethical Compliance

QASPER - Dataset Identity

Source

AllenAI - allenai/qasper on HuggingFace

888 Papers | 2,321 QA Pairs

NLP research papers (train split). Answer types: Extractive 56.6% | Abstractive 26.3% | Yes/No 17.1%

Structure

Full JSON with section titles and paragraph boundaries. Ground-truth evidence paragraphs per question - essential for RAGAS Context Recall computation.

Licence

CC BY 4.0 - free for research use with attribution. No restrictions on publishing derived results.

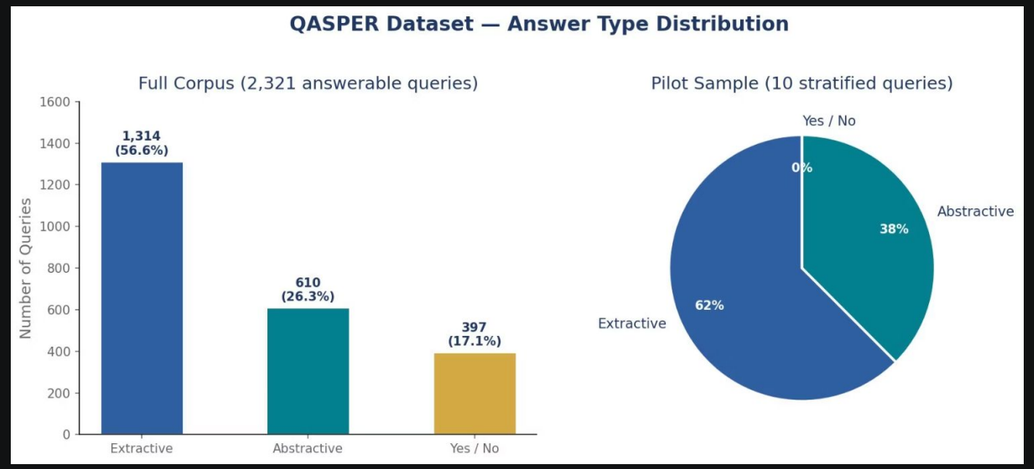
Why QASPER Stresses Naive RAG

Questions are written from abstracts, but answers require evidence from the full body text - adversarial to top-K retrieval. NLP papers share vocabulary (attention, transformer, embedding) - a perfect condition for triggering retrieval noise.

Ethical Compliance

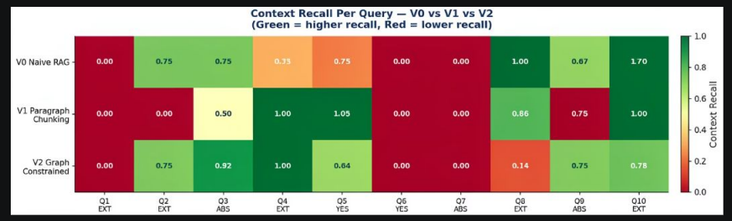
No personal data - published academic papers only. No anonymisation required. CC BY 4.0 licence explicitly permits research use with attribution. No UH Ethics approval required. Fully documented origin, licence, and collection methodology.

EDA and Experimental Design - QASPER Corpus



Answer Type Distribution

The corpus is extractive-majority at 56.6% - 1,314 out of 2,321 answerable queries require exact evidence spans from within the paper body. This matters because extractive questions are precisely the ones where fixed-window chunking causes the most damage - the evidence span exists in the corpus but gets split across a chunk boundary. The pilot sample mirrors this distribution at 62% extractive, 38% abstractive, 0% yes/no, ensuring no variant benefits from an easier query mix.



Context Recall Per Query - V0 vs V1 vs V2

Read it row by row. The V0 row shows mixed performance - strong on Q2, Q3, Q8, Q10 but failing completely on Q1, Q6, Q7. The V1 row shows the unexpected drop - Q2 falls from 0.75 to 0.00, Q9 drops from 0.67 to 0.00. This is the finer granularity effect - more chunks, more competition, correct paragraphs getting outranked. The V2 row recovers - Q2 back to 0.75, Q3 up to 0.92, Q9 up to 0.75. The one exception is Q8 where V2 drops sharply to 0.14. That is the anchor noise case - top-1 retrieved from the wrong paper, graph expanded from a bad anchor and amplified the error rather than correcting it. That single failure is the key limitation of the current V2 design.

EDA - Corpus Insights and Design Rationale

Extractive Majority

56.6% of answers are extractive - evidence spans confirm that fixed-window chunking splits are the primary Context Loss vector in baseline retrieval. The corpus distribution directly motivates the structural chunking intervention in V1.

Stratified Sampling

The stratified sample locks query distribution across V0, V1, and V2. This ensures metric differences reflect genuine pipeline improvements, not query mix variance. RANDOM_SEED=42 guarantees the same 10 queries are used across all three variants.

Heatmap Analysis

Per-query patterns reveal V2 improves most extractive queries. The Q8 drop is confirmed as anchor noise - top-1 retrieved from the wrong paper, causing the graph to expand from a bad anchor.

V2 row shows strongest recall on extractive queries (Q2, Q3, Q4, Q10). Q8 drop confirmed as anchor noise - top-1 retrieved from wrong paper; graph expanded from bad anchor.

Methodology - Three Pipeline Variants

Step 1 - V0: Naive RAG Baseline

Fixed-window chunking (256 tokens), top-5 cosine similarity, no structural awareness. Measures failure mode incidence (Context Loss + Retrieval Noise).

Step 2 - V1: Structure-Aware Chunking

Paragraph-boundary chunking (variable length), top-5 cosine similarity. Isolates chunking effect on Context Loss. V0 → V1 measures chunking impact.

Step 3 - V2: Graph-Constrained Retrieval

Paragraph chunking + K=10 initial retrieval, 2-hop graph proximity filter, adjacency expansion and provenance tracking. Isolates graph constraint effect on Retrieval Noise. V1 → V2 measures retrieval constraint impact.

Each variant is constructed to isolate a single independent variable: V0 → V1 measures the chunking impact on Context Loss; V1 → V2 measures the retrieval constraint impact on Retrieval Noise. All other components remain fixed across variants.

Fixed across all variants: Embedding model (all-MiniLM-L6-v2) | LLM (Llama 3.2 3B, temp=0) | FAISS IndexFlatL2 | Query set (queries.json) | Prompt template

Technical Stack - Models and Tools

all-MiniLM-L6-v2

Embedding Model. Compact enough to embed 46,882 paragraphs in <100MB RAM. Well-benchmarked on BEIR retrieval tasks. Fast CPU inference (~14ms/sentence). Held constant - absolute quality less important than consistency across variants.

FAISS IndexFlat 2

Vector Index. Exact search guarantees reproducible top-K results. Approximate indexes (e.g. HNSW) introduce approximation error - results may omit true top-K neighbours. HNSW is deterministic with a fixed seed, but approximation error is non-zero and non-reproducible across index builds. Fully deterministic, zero approximation error.

Llama 3.2 3B via Ollama

LLM. Runs locally on M1 Mac 8GB - zero API cost, no rate limits. Temperature=0 makes generation fully deterministic. Held constant across V0/V1/V2 so generation quality is a fixed confound, not a variable.

NetworkX

Graph Library. ~59K nodes fit in RAM with zero infrastructure overhead - no Neo4j server required. BFS and shortest path built-in. Built once, serialised to a 26MB pickle, loaded in under 1 second.

RAGAS

Evaluation Framework. Component-level metrics separate retrieval failures from generation failures - standard QA metrics (EM, F1) cannot attribute failure to a specific pipeline stage. Provides retrieval-phase visibility essential for failure-mode-disaggregated evaluation.

V2 - Graph Design and 3-Step Retrieval Constraint

59,300

Total Nodes

46,882 paragraph nodes + 12,418 virtual section nodes

81,346

Total Edges

Sequential adjacency + section membership edges

26MB

Graph Size

Serialised to pickle - loads in under 1 second at evaluation time

1

Widened Vector Search

Retrieve K=10 candidates (not K=5) to give the graph filter sufficient candidates to work with during re-scoring.

2

Anchor + Graph Re-scoring

Top-1 chunk = anchor node. Score remaining 9 by hop distance. Within 2 hops → +0.3 bonus. No path → ×0.5 penalty. Select final top-5.

3

Adjacency Expansion

If the anchor's sequential neighbours did not appear in the initial top-10, add them. Directly targets Context Loss - recovers the paragraph that vector search missed.

V2 - Graph Construction and Re-Scoring: Code Implementation

Each paragraph becomes one node with metadata (paper ID, section name, paragraph index). Two edge types - sequential adjacency for local continuity and section membership for hierarchical structure - enable both local and global graph queries.

Snippet 1 - Graph Construction

```
# Each paragraph = one node
G.add_node(chunk["chunk_id"],
    paper_id=chunk["paper_id"],
    section_name=chunk["section_name"],
    paragraph_index=chunk["paragraph_index"])

# Edge Type 1 - sequential adjacency
G.add_edge(para_N, para_N1,
    type="sequential")

# Edge Type 2 - section membership
G.add_edge(chunk_id, virtual_id,
    type="section")
```

Snippet 2 - 2-Hop Proximity Re-scoring

```
# Top-1 = anchor node
anchor_id = candidates[0]["chunk_id"]
for c in candidates[1:]:
  try:
    hops = nx.shortest_path_length(
        G, anchor_id, c["chunk_id"])
    if hops <= HOP_LIMIT:  # 2
      c["score"] += PROXIMITY_BONUS  # +0.3
    else:
      c["score"] *= PENALTY  # ×0.5
  except nx.NetworkXNoPath:
    c["score"] *= PENALTY  # disconnected
```

Graph is built once during pre-processing (0.2s build time) and serialised to pickle. `nx.shortest_path_length` uses BFS - returns integer hop count or raises `NetworkXNoPath` for structurally disconnected chunks. Inter-paper edges are absent by design, so chunks from different papers are always penalised.

Evaluation Metrics - Four-Layer Design

No single metric is sufficient. Context Loss and Retrieval Noise fail at different pipeline layers - each requires instrumentation at its own layer.

Context Recall - Retrieval Layer

Decomposes ground-truth answers into atomic claims. Checks each claim against retrieved chunks. Score = claims covered / total claims. **Targets Context Loss.** Applied to V0, V1, V2. Threshold: **< 0.70 = failure.**

Context Precision - Retrieval Layer

Measures proportion of retrieved chunks that are genuinely relevant to the query. Catches noisy chunks at chunk level - before generation reaches the LLM. **Targets Retrieval Noise.** Applied to V0, V1, V2. Threshold: **< 0.70 = failure.**

Faithfulness - Generation Layer

Extracts claims from the generated answer and verifies each against retrieved context. Measures answer grounding - but cannot detect noise if the answer sounds grounded in wrong evidence. Threshold: **< 0.75 = failure.**

Provenance Coverage - Pipeline Layer

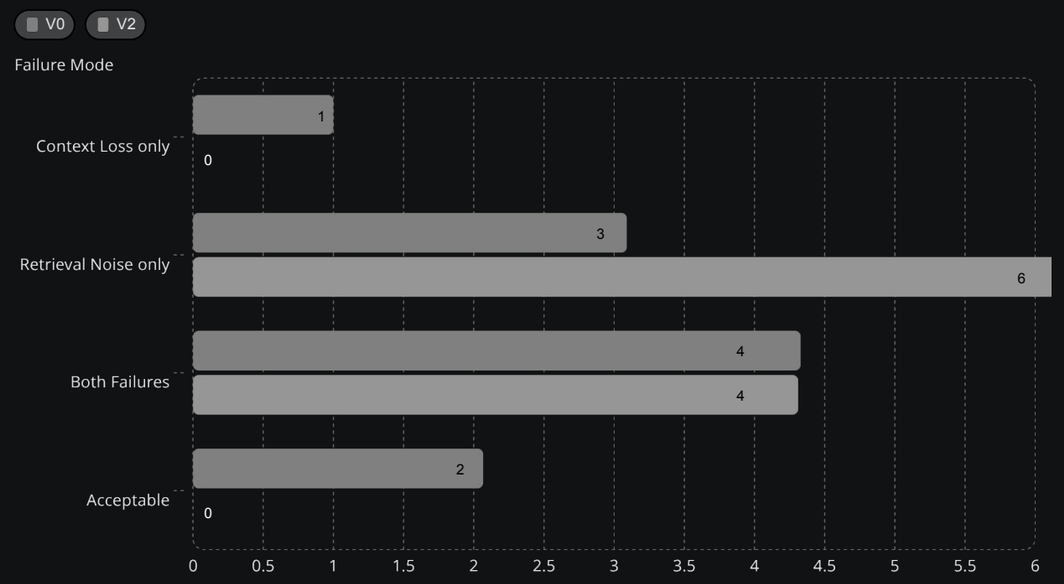
Measures % of retrieved chunks with a traceable graph path to the anchor node within 2 hops. Operationalises explainability as a computable score. **Novel metric defined for this study.** Applied to V2 only. Descriptive - higher = better.

Results - Recall Progression and Failure Mode Shift

Failure Mode Classification - V0 vs V2 (n=10 pilot queries)

Failure Mode	V0 Count	V2 Count
Context Loss only	1 of 10	0 of 10
Retrieval Noise only	3 of 10	6 of 10
Both Failures	4 of 10	4 of 10
Acceptable (no failure)	2 of 10	0 of 10

n=10 pilot - raw counts only. Percentages not reported; sample size is insufficient for statistical inference.



Raw counts only - not proportions. n=10 pilot.

Results - Key Findings

V2 Recall 0.498 > V0 0.475

Graph adjacency expansion recovers evidence that paragraph chunking alone missed. The structural constraint performs real work beyond simple re-ranking.

Context Loss Reduced (Pilot)

Pilot suggests reduction in isolated Context Loss cases - 1 of 10 queries in V0 showed isolated Context Loss; 0 of 10 in V2. Confirmation pending 150-query HPC evaluation. Paragraph-boundary chunking paired with graph expansion appears sufficient for this failure mode.

Retrieval Noise Persists (6 of 10 queries)

The dominant remaining failure mode - present in 6 of 10 V2 queries vs 3 of 10 in V0. Note: n=10 pilot only; counts are illustrative, not statistically generalisable. Full 150-query HPC evaluation pending.

Pilot Results - V0 vs V1 vs V2 (10 Queries)

Metric	V0 Naive RAG	V1 Paragraph	V2 Graph	Threshold
Context Recall	0.475	0.436	**0.498** ▲	< 0.70 = fail
Context Precision	0.270	-	-	< 0.70 = fail
Faithfulness	0.083 ⚠️	-	-	< 0.75 = fail
Provenance Coverage	N/A	N/A	**0.480**	Descriptive

Context Precision and Faithfulness for V1 and V2 are pending full HPC evaluation with the Llama 3.1 8B judge model. V0 baseline scores confirm both failure modes - precision at 0.270 and faithfulness at 0.083 fall well below the defined thresholds, validating the motivation for structural intervention.

V0 Faithfulness score unreliable - Llama 3.2 3B produced JSON output failures during RAGAS evaluation; score should be treated as indicative only. Pending HPC validation with Llama 3.1 8B judge.

Analysis - Results Interpretation and Limitations

V1 Recall Drop (0.475 → 0.436)

Paragraph chunks are shorter (mean 65 tokens vs 250) and more numerous (46,882 vs 22,165). The correct evidence paragraph has greater competition in the index. Finer granularity increases retrieval difficulty without structural filtering to compensate.

V2 Recovery (0.436 → 0.498)

Graph adjacency expansion recovers what V1 lost. When vector search misses the correct paragraph, the graph pulls in its sequential neighbour. The structural constraint is doing real work - not merely re-ranking the same results.

Anchor Noise Limitation

When top-1 is a noisy anchor from the wrong paper, the graph filter expands from a bad starting point and may amplify the error. Observed in Q8 pilot. V2 assumes the anchor is correct - a key known limitation.

Hyperparameter Design

Parameter	Value	Justification
TOP_K_INITIAL	10	Wider net for graph re-scoring
TOP_K_FINAL	5	Consistent context size to LLM
HOP_LIMIT	2	Balances recall recovery against anchor noise amplification risk - covers sequential + section neighbours without over-expanding from a noisy anchor.
PROXIMITY_BONUS	+0.3	Structurally adjacent reward
PENALTY	×0.5	Structurally disconnected penalty

Conclusions and Next Steps

> Graph-based structural constraints measurably improve Context Recall (V0: 0.475 → V2: 0.498) while Provenance Coverage (0.48) provides a novel quantitative explainability signal - with no knowledge graph extraction overhead.

What Was Built

01

V0 Naive RAG Baseline

Failure modes confirmed - recall 0.475, precision 0.270

02

V1 Structure-Aware Chunking

Paragraph boundaries preserved; finer retrieval granularity established

03

V2 Graph-Constrained Retrieval

2-hop filter + adjacency expansion + provenance tracking

04

Full RAGAS Evaluation Pipeline

Running locally on M1 Mac; HPC-ready for full 150-query run

Next Steps

HPC Cluster (V100/A100)

Full 150-query evaluation across all three variants

Larger RAGAS Judge

Llama 3.1 8B - reliable Precision and Faithfulness scores

Ablation Study

HOP_LIMIT and PROXIMITY_BONUS sensitivity analysis

Final Project Report

Dissertation write-up incorporating full evaluation results

Wider Implications:• Practitioner: Lightweight structural constraints are sufficient - no full Graph RAG complexity is needed to address common failure modes in production RAG systems.

• Research: Graph-based proximity filtering is a viable, low-overhead alternative to full knowledge graph extraction for improving retrieval quality.

• Provenance Coverage (0.48): A novel quantitative explainability signal - structural traceability is measurable without additional annotation overhead.

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